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SUPERSEDING
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MILITARY SPECIFICATION

AIR-FILTER MEDIUM

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of air-filter medium made of processed wool felt for use in filtering particulate materials in the form of dusts.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CCC-T-191 — Textile Test Methods.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Standards

D461—Standard Methods of Testing Felt.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

UNITED STATES BUREAU OF MINES

Schedule 21—Procedure for Testing Filter-Type Dust, Fume, and Mist Respirators for Permissability.

(Application for copies should be addressed to United States Bureau of Mines, Washington, D. C.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

FSC 4240

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3. REQUIREMENTS

3.1 Manufacture. The air-filter medium shall be manufactured from 100-percent wool felt which has been impregnated by immersion in a solution containing a zinc-hardened rosin, dried, and then mechanically processed to produce a medium having electrostatic filtering properties.

3.2 Felt. The felt used in the air-filter medium shall be of the grade known commercialy as "first", and shall be a fabric built up by the interlocking of fibers by a suitable combination of mechanical work, chemical action, moisture, and heat, without spinning, weaving, or knitting. The felt shall be limited to what is commonly referred to as wool felt and shall not include hair; punched, woven, stitched, or quilted felt; paper, or other materials of felt-like appearance, which are products of entirely different construction and properties.

3.3 Wool. The wool used in the felt shall be of fleece, pulled sheep wool, noils, reprocessed wool or any combination of these materials conforming to U. S. standards of grades used in commercial practice in the manufacture of wool felt. Wool known commercially as "coarse Mexican" shall not be used.

3.4 Chemical and physical properties. The chemical and physical properties of the air-filter medium shall conform to the requirements stipulated in table I when tested as specified in the test paragraphs referenced therein.

TABLE I. *Chemical and Physical Properties*

Property	Requirement	Test paragraph
Thickness	Nominal — 0.080 in. Min — 0.065 in. Max — 0.100 in.	4.5.2
Wt per sq yd	Min — 11.5 oz Max — 15.0 oz	4.5.3
* Fiber content	100% wool	4.5.4
* CCl_4 soluble material	Max — 1.50% (by wt)	4.5.5
Rosin content	Min — 15% (by wt) Max — 30% (by wt)	4.5.6
Breaking strength	Min — 30 lbs (2 in. wide strip)	4.5.7
Initial Resistance to airflow	Max — 0.50 in water (gage) at airflow of 85 liters per minute	4.5.8
Total amount of high-type A dust-unretained after concentration test.	Max — 9.0 mg (for three test specimens) Max — 4.0 mg (for any one test specimen)	4.5.9
Resistance to airflow after high-type A dust-concentration.	Max — 2.00 in. water (gage) at airflow of 85 liters per minute	4.5.10

* Of untreated felt.

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3.5 **Workmanship.** The air-filter medium shall be free from thick and thin spots, splices, abrasions, tears, punctures, wrinkles, dirt inclusions, and other imperfections which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the inspection procedures of this specification shall govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.3, 3.5 and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the air-filter medium produced by one manufacturer, from one machine, using one bath of treating solution, and under essentially the same manufacturing conditions. However, not more than one lot of wool felt shall be included in any one lot of finished air-filter medium.

4.4 **Sampling for tests.**

4.4.1 *Untreated felt.* Sufficient untreated felt shall be selected from each lot and prepared for testing as specified in 4.5.1.2 and 4.5.1.3.

4.4.2 *Air-filter medium.* Sufficient air-filter medium shall be selected from each lot and prepared for testing as specified in 4.5.1.1, 4.5.1.4, 4.5.1.5, and 4.5.1.6.

4.5 **Tests.**

4.5.1 *Specimens.*

4.5.1.1 *For thickness and weight tests.* Three test specimens each containing not less than 40 square inches (sq. in.) shall be cut from the air-filter medium sample (see 4.4.2) for thickness and weight tests.

4.5.1.2 *For fiber content test.* Two to five grams (g) shall be cut from the untreated felt sample (see 4.4.1) for the fiber content test.

4.5.1.3 *For soluble material in carbon tetrachloride test.* Approximately 5 g shall be cut from the untreated felt sample for the soluble material in carbon tetrachloride test.

4.5.1.4 *For rosin content test.* Approximately 5 g shall be cut from the air-filter medium sample for the rosin content test.

4.5.1.5 *For breaking strength test.* Five strips, each 2 in. wide and not less than 6 in. long, shall be cut from the air-filter medium sample in accordance with method 5102 of Specification CCC – T – 191 for the breaking strength test.

4.5.1.6 *For initial resistance airflow test, high-type A dust-concentration test, and resistance to airflow after high-type A dust-concentration test.* Three circular test specimens each having a test area of 10 sq. in. shall be cut from the air-filter medium sample for the initial resistance to airflow test, for the high-type A dust-concentration test,

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and for the resistance to airflow after high-type A dust-concentration test.

4.5.2 *Thickness.* The thickness of the three test specimens of air-filter medium (see 4.5.1.1) shall be measured by a dial micrometer. The pressure foot shall have an area of 1 sq. in. (1.129 in. dia.) and shall be actuated by a freely acting dead load of 10 ounces. The load shall be applied gradually and without shock. The thickness reading shall be taken at least 10 seconds after the load is applied. A minimum of five measurements evenly distributed over the specimen shall be made and the average considered the thickness of the felt.

4.5.3 *Weight.* Following the thickness test, the three test specimens shall be accurately measured, and then weighed on a balance having a sensitivity of 0.1 ounce. The average weight and area shall be used in calculating the weight per square yard.

4.5.4 *Fiber content.* The fiber content of the test specimen of untreated felt (see 4.5.1.2) shall be determined in accordance with method 2101.1 of Specification CCC–T–191.

4.5.5 *Soluble material in carbon tetrachloride.* The test specimen of untreated felt (see 4.5.1.3) shall be cut into ⅛-in. squares, dried to constant weight at 105° to 110°C, and the weight recorded. The specimen shall be extracted with carbon tetrachloride in a soxhlet apparatus for 4 hours. The carbon tetrachloride shall then be evaporated from the extract, and the residue dried to constant weight at 105° to 110°C. (This residue is matter soluble in carbon tetrachloride and shall be reported as a percentage of the weight of the original oven-dried specimen.)

4.5.6 *Rosin content.* The rosin content of the test specimen of air-filter medium (see 4.5.1.4) shall be determined by repeating the test specified in 4.5.5 (except air-filter medium is used instead of untreated felt) and subtracting from the result, the soluble material in carbon tetrachloride.

4.5.7 *Breaking strength.* The breaking strength of the five test specimens of air-filter medium (see 4.5.1.5) shall be determined in accordance with method 5102, Specification CCC–T–191.

4.5.8 *Initial resistance to airflow.* The initial resistance to airflow of the three circular test specimens of air-filter medium (see 4.5.1.6) shall be determined in accordance with ASTM Standard D461. (The airflow rate shall be 85 liters per minute.)

4.5.9 *High-type A dust-concentration test.* The high-type A dust-concentration test shall be performed on the three test specimens of air-filter medium (see 4.5.1.6) in accordance with United States Bureau of Mines Schedule 21. (The result of this test is the total amount of high-type A dust unretained.)

4.5.10 *Resistance to air-flow after the high-type A dust-concentration test.* The resistance to airflow after the high-type A dust-concentration test of the three circular test specimens (see 4.5.1.6) of air-filter medium shall be determined in accordance with ASTM Standard D461. (The air-flow rate shall be 85 liters per minute.)

4.6 Acceptance/rejection criteria. If the sample air-filter medium or untreated felt when tested as specified in 4.5.2 to 4.5.10, inclusive, fails to meet the applicable requirement specified in 3.4, the lot represented lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. Packaging shall be in accordance with the manufacturer's commercial practice to provide adequate protection for the items packed as specified in 5.2.

5.2 Packing, level C. A uniform quantity of air-filter medium, packaged as specified in 5.1, shall be packed in a manner to insure

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carrier acceptance and safe delivery at destination. Shipping containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Army—CmlC
Navy—MC
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Review interest:
Army—C, E,
Navy—MC
Air Force—WRA

User interest:
Army— M, O,

Preparing activity:
Army—CmlC

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